

a

A traceable gene-to-cell contract defines each annotation

The visual hierarchy begins with the data contract, not a headline score



— frozen corpus

272,732 cells

— strict transfer

3,964 cells

— adaptation registry

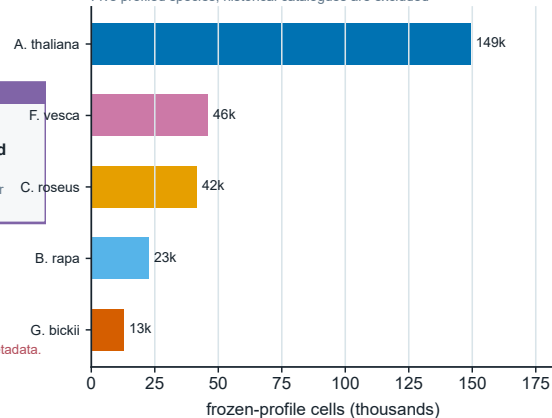
24 species modules

All reported outputs retain mapping, denominator and scope metadata.

b

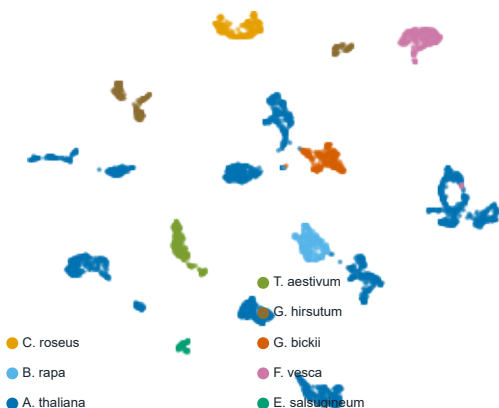
Training profile is compact and declared

Five profiled species; historical catalogues are excluded



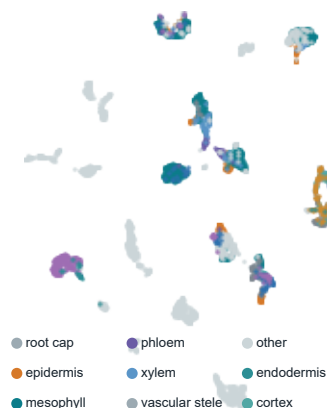
c Held-out species occupy a shared representation

3,964 cells; color encodes species, not predicted identity



d Ontology states recur across species

Recorded separately from species identity



e Frozen species-by-tissue matrix

Only profile cells enter this matrix

